

Samay — a court day that runs to plan

PUCAR "Scheduling Justice" hackathon · FOSS United Week · 24 September 2026

Team: samay · **Members:** Gaurav (Guruprasad) Meena, *add teammates* · **Level claimed:** L3

Code: submissions/samay/ on branch team-samay · **Run it:** `pip install -r requirements.txt && streamlit run app.py`

1. The problem in one line

A cheque-bounce (s.138 NI Act) judge lists ~60 cases a day, reaches about a third, and only ~9 hearings actually move a case forward; the rest are adjourned and handed a flat 60-day date. Old cases wait, litigants make wasted trips, and nobody knows when their case will be called.

2. What Samay does

Every evening, for one judge, Samay decides **which** cases to list tomorrow, **in which sitting and at what time**, and **the next date** for every case that does not move — using only the court's own data.

Step	What happens	Where
1. Intake	Court master uploads the judge's docket (Excel/CSV). Header spellings, dd/mm dates, blank or duplicate rows and multi-sheet workbooks are handled; wrong files get a clear message.	src/intake.py
2. Read every case	The last order is read: what happened, what the case is waiting on, who was absent, "last chance", first vs repeat hearing at this stage.	src/orders.py, src/model.py
3. Score 0–100	The team's Case Priority Scoring algorithm, implemented exactly (section 3).	src/score100.py
4. Hold back	Cases whose summons / warrant / notice is still out are Conditional and not listed (rule R1).	src/model.py
5. Build the day	Bail first (liberty lane) → cases overdue 30+ days → a protected share for 4+ year cases → the rest by score × P(moves forward)^{1.5} ÷ expected minutes , packed into 10:45–12:30 and 13:30–17:00 with a 2-min changeover and a start time per case.	src/priority.py, src/packer.py
6. Readiness check	Two days before, advocates confirm; "need time" frees the slot for the next case — no wasted trip.	src/simulate.py
7. Next date	By <i>why</i> it didn't move: court didn't sit → next sitting day; party absent → sooner; summons pending → when it's expected back. Never a flat 60 days. Holidays and the judge's leave respected.	src/next_date.py
8. Judge approves	Judge sees the list, removes or adds cases, approves; the causelist downloads.	ui_pages/judge.py

3. Case priority score — exactly as specified

$$\text{Priority} = 35 \times \text{age_value} + 25 \times \text{readiness_value} + 15 \times \text{disposal_value} + 15 \times \text{churn_value} + 10 \times \text{urgency_value}$$

- **Age** — years since filing ÷ full-points age (mean + 2 sd of the roster = **8 years**, fixed once).
- **Readiness** — P(hearing of this type makes progress) from `substantiveness_by_hearing_type.csv` × attendance factor (0.7–1.0 from who the hearing needs vs who was present).
- **Disposal proximity** — position of the current stage in the 11-stage lifecycle.

- **Churn** — hearings held ÷ median hearings expected by this stage (hearing_type_reference.csv), full points at 2×.
- **Urgency** — "last chance" 1.0, "for judgment" 0.8.

Rules beside the score: R1 Eligible/Conditional, R2 age weight never below 20, R3 bail liberty lane, R4 Backlog 4y+ / Ageing 5y+ flags, R7 no case waits forever (overdue forced in).

Verified on the real data: on all 100 cases in roster_sample_100.csv, Samay reproduces the document's score, status and flags for **100 / 100 cases** (range 7.2–91.3, mean 37.7, median 35.2; 80 Eligible, 20 Conditional). Example: ST/1261/2017 = 35.0 + 25.0 + 13.5 + 9.8 + 8.0 = **91.3**.

4. Results (60 sitting days from 24 Sep 2026, same simulator for both)

On the real 100 cases (data/roster_sample_100.csv):

	Today's court	Samay
Listed cases reached	49%	97%
Heard hearings that move the case	63%	91%
Hearings that move a case, per day	0.5	4.3
4+ year cases heard at least once	30%	87%
5+ year cases moved a stage	13%	73%
Started within 30 min of the slot	12%	81%
Next date sensible for the case	12%	90%
Cases disposed	5	54

At the case study's scale (3,000 cases resampled from the real 100):

	Today's court	Samay
Listed per day	60	21
Hearings that move a case, per day	8.8	14.2 (+62%)
Listed cases reached	35%	86%
Wasted trips in 60 days	2,856	378 (-87%)
5+ year cases moved a stage	12%	43%
Cases disposed	135	437 (3.2×)

Fewer listings, far more progress: Samay lists what is ready and fits, instead of listing everything and adjourning most of it.

5. The prototype (one command: `streamlit run app.py`)

- **Sign in** as a judge or the court master.
- **Judge:** *Today* (timeline + causelist with time, listing number, score, why listed; remove/approve/download) · *Calendar* (month view, holidays, leave) · *Cases* (score, status, flags, last order, journey) · *Priority* (re-weight live; age floor 20) · *Insight* (Samay vs today's rules) · *How Samay decides*.
- **Court master:** *Judges* (each bench's numbers) · *Files* (upload a docket, set leave) · *Run a day* (record outcomes; next date follows the rule) · *New complaint* (s.138 limitation and papers check before listing).

- **Analytics dashboard:** what-if levers, three judges' rules compared, why hearings fail, advocates (L3), rankers.

6. Complexity: L3

- **L2:** sampled durations, start times and changeovers; per-type outcome and failure-reason distributions; prerequisites that change over time; per-case attendance and preparedness.
- **L3:** each advocate is an agent (diligent / overloaded / dilatory) who learns from the court's incentives (reminders, cost for on-the-day adjournment) — see *Advocates (L3)* and `agents_study.py`.

7. Data and honesty

- Only the organisers' files in `data/` are used: roster, hearing reference, substantiveness, failure reasons, court calendar. No invented cases in the prototype — the judge's default docket is the real 100 cases.
- The 3,000-case run resamples the real 100 (`scripts/generate_roster.py`) only to show scale.
- Fixed rules (0.7–1.0 attendance range, "last chance" = 1.0, 30-day overdue, 2-min changeover, 10:30–11:00 start, lunch 12:30–13:30) are named constants, documented in `case_priority_scoring.md` and `src/config.py`.
- Limits: one case type (s.138), one judge at a time, simulated outcomes (not yet validated on live court results).

8. Repository

`app.py` (entry) · `ui_pages/` (sign-in, judge, court master, dashboard) · `src/` (engine: intake, orders, model, score100, priority, packer, simulate, next_date, agents, metrics) · `tests/` (8 passing) · `samples/` (Excel template) · `results.md`, `rankers.md`.